SG No. 165 D_{3d}^4 $P\bar{3}c1$ [trigonal]

* plus set: $+[0,\ 0,\ 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `32.`

No.	position	mapping
1	$\left[0,\ 0,\ \frac{1}{4}\right]$	`[1,2,3,4,5,6]`
2	$\left[0,\ 0,\ \frac{3}{4}\right]$	`[7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-3..`

No.	position	mapping
1	$[0,\ 0,\ 0]$	`[1,2,3,7,8,9]`
2	$\left[0,\ 0,\ \frac{1}{2}\right]$	`[4,5,6,10,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `3..`

No.	position	mapping
1	$[0,\ 0,\ z]$	`[1,2,3]`
2	$\left[0,\ 0,\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$[0,\ 0,\ -z]$	`[7,8,9]`
4	$\left[0,\ 0,\ z+\frac{1}{2}\right]$	`[10,11,12]`

Table 4: Wyckoff site: `4d`, site symmetry: `3..`

No.	position	mapping
1	$\left[\frac{1}{3},\ \frac{2}{3},\ z\right]$	`[1,2,3]`
2	$\left[\frac{2}{3},\ \frac{1}{3},\ \frac{1}{2}-z\right]$	`[4,5,6]`
3	$\left[\frac{2}{3},\ \frac{1}{3},\ -z\right]$	`[7,8,9]`
4	$\left[\frac{1}{3},\ \frac{2}{3},\ z+\frac{1}{2}\right]$	`[10,11,12]`

Table 5: Wyckoff site: `6e`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{2},\ 0,\ 0\right]$	`[1,7]`
2	$\left[0,\ \frac{1}{2},\ 0\right]$	`[2,8]`
3	$\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$	`[3,9]`
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$	`[4,10]`
5	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$	`[5,11]`
6	$\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$	`[6,12]`

Table 6: Wyckoff site: `6f`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	`[1,5]`
2	$\left[0,\ x,\ \frac{1}{4}\right]$	`[2,4]`
3	$\left[-x,\ -x,\ \frac{1}{4}\right]$	`[3,6]`
4	$\left[-x,\ 0,\ \frac{3}{4}\right]$	`[7,11]`
5	$\left[0,\ -x,\ \frac{3}{4}\right]$	`[8,10]`
6	$\left[x,\ x,\ \frac{3}{4}\right]$	`[9,12]`

Table 7: Wyckoff site: `12g`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-y,\ x-y,\ z]$	`[2]`
3	$[-x+y,\ -x,\ z]$	`[3]`
4	$\left[y,\ x,\ \frac{1}{2}-z\right]$	`[4]`
5	$\left[x-y,\ -y,\ \frac{1}{2}-z\right]$	`[5]`
6	$\left[-x,\ -x+y,\ \frac{1}{2}-z\right]$	`[6]`
7	$[-x,\ -y,\ -z]$	`[7]`
8	$[y,\ -x+y,\ -z]$	`[8]`
9	$[x-y,\ x,\ -z]$	`[9]`
10	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	`[10]`
11	$\left[-x+y,\ y,\ z+\frac{1}{2}\right]$	`[11]`
12	$\left[x,\ x-y,\ z+\frac{1}{2}\right]$	`[12]`